

In my Cycle: Designing Sensory Spaces for Menstrual Awareness through Soma Design

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Abstract

This demonstration presents *In My Cycle*, an interactive installation that externalizes hormonal and emotional sensations through touch, sound, and spatial storytelling. Developed through artistic performance and soma design, the work translates lived menstrual experience into a walkable sound installation consisting of 19 suspended wooden spheres. Each sphere represents one recurring bodily sensation encountered across two decades of menstrual cycles. Through quiet interaction and embodied memory, the installation creates a shared space for self-reflection, conversation, and menstrual literacy. By engaging with feminist HCI, soma design, and critical menstruation studies, this work explores how menstrual technologies can become sites of poetic attention, public dialogue, and intimate knowledge exchange.

CCS Concepts

• Human-centered computing → Interaction design process and methods.

Keywords

First-person research; soma design; menstrual health; informal learning, artistic research

ACM Reference Format:

Lisa Hofer. 2025. In my Cycle: Designing Sensory Spaces for Menstrual Awareness through Soma Design. In *Designing Interactive Systems Conference (DIS '25 Companion)*, July 05–09, 2025, Funchal, Portugal. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3715668.3735598>

1 Introduction

In this work, I explore a personal and embodied design practice that aims to make the inner self, particularly hormonal and emotional sensations, visible, tangible, and audible. Through the translation of menstrual experiences into a walkable and interactive installation, I investigate how technology and design can externalize internal bodily states in ways that are both personally meaningful and socially engaging. Menstruation is often associated with invisibility through cultural concealment, shame, or medical abstraction [1, 9, 14]. Critical menstruation studies reveal how stigma and disembodiment shape menstrual experiences, calling for social change, education, and inclusive design [1, 5, 9]. Research at



Figure 1: *In my Cycle*

the intersection of menstrual health and HCI draws from feminist interaction design, critical menstruation studies, and embodied computing. Scholars have explored how technology can move beyond tracking to challenge stigma, support bodily literacy, and affirm diverse experiences [5, 13, 16]. Digital menstrual technologies tend to prioritize prediction, datafication, and optimization, overlooking the sensory, affective, and lived dimensions of the cycle [6, 16]. This work offers an alternative: a soma design approach, through performance, storytelling, embodied memory and informal learning. Focusing on the externalization of sensations, I am using methodologies of soma design, by positioning my own menstruating body at the centre and by applying an *autosomatographical* [7], and autobiographical approach [15]. I engage with the question of how menstrual health can be more understandable through embodied performances and physical forms. The performance part is a way of mapping my own experiences on and with the body. Furthermore, the sensory artefact expresses internal experiences using sound and touch which serves as a tool for self-reflection, communication, and



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ACM ISBN 979-8-4007-1486-3/25/07

<https://doi.org/10.1145/3715668.3735598>

public engagement. Due to its scale and form factor, the artefact itself forms a physical environment, a space within a space, that fosters knowledge exchange and raises awareness about a subject that urgently needs greater visibility in public discourse [3].

2 Methodological Background

This project was twofold and started at an art and maker festival in September 2023 with a performance. One year later, in September 2024, I developed an installation as a further iteration of my conducted research at the same festival. The artistic and soma design-driven artefacts build up on each other and I will explain the process, the methods and the methodologies used here.

For the performance, I combined artistic research with a soma design approach and used body mapping. Artistic research engages with ambiguity, emotion, aesthetics, and the sensory body [4]. It creates situated, experiential forms of knowledge that challenge dominant, rationalist paradigms in technology design [15]. HCI refers to the use of artistic practices, such as performance, installation, visual art, or sound, as both a method and outcome of research [8]. Soma design involves engaging with bodily sensations, often through movement, touch, or reflection [10]. Different methods e.g. body maps, body storming, autobiographical and autosomatographical documentation are used to get closer to the lived experiences. My work builds on the externalization of felt sensations and I am using autobiographical and autosomatographical methods to understand and translate lived experience into touch- and sound-based extensions that foster self-awareness and public dialogue. Autobiographical design is used by involving the own life, body, and lived experience as the research context, which can be useful for intimate, stigmatized, or embodied topics (e.g., menstruation, chronic illness) [12]. Autosomatography emerged from disability studies and is used in chronic pain research. In HCI research autosomatography captures multisensory, embodied self-reporting of lived experience [7]. It resonates well with menstruation sensations as an intimate and often invisible experience. For the installation, I made use of storytelling and tangible materials as I focused on the externalization of sensations. The use of tangible interaction technology and sharing stories have proven effective in promoting menstrual literacy and normalizing discussion [11, 16]. Grounded in personal narratives, the translation of feelings serves as reflective tools that externalize sensation and experience [7]. This artistic practice is not used simply to communicate research but as the method through which knowledge is generated [8].

3 Design

3.1 Design Process

20x12. In order to gain a deeper understanding of my body I enacted an autosomatographical process under the name *20x12: A Body-Mapped Archive of Menstrual Memory*. For this process, I marked 240 red ink dots across my body, each symbolizing a month of bleeding over 20 years. With each mark, I recalled where I had felt the pain, fatigue, and confusion, both physically and mentally. I identified 19 recurring sensations that had shaped my experience of the cycle, not only during bleeding but throughout the hormonal rhythm and how it affected my entire body and mental state. This autosomatographical performance became a way



Figure 2: hanging and moving spheres

of making the invisible visible. I treated the body as a lived body map, an archive of physical and emotional memory. Each gesture, mark, or movement was a means of accessing internal knowledge that had long gone unspoken. It also revealed the emotional weight of cultural concealment, internalized shame, and misunderstanding: sensations I had once misunderstood as laziness or weakness were hormonal signals I did not know how to read. Through this process, I arrived at the insight that I was not weak or lazy, I was hormonally incapable. This realization became the conceptual and emotional seed for the interactive installation *In My Cycle*.

3.2 Installation

In My Cycle. The 19 sensations identified in the performance became the foundation for the design of the installation. *In My Cycle* is an embodied, interactive system that expresses inner states and cyclical knowledge. It consists of 19 spheres suspended from the ceiling in a circular formation (Fig.1). It creates a calm environment in which to encounter the rhythm of my hormonal cycle. The wooden spheres are half coloured half not. The wooden spheres should represent the raw, grounded materiality of the body (Fig.5). The silver-painted spheres are intended to reflect the social process of making the invisible visible. All spheres are connected through an elastic conductive thread with two bare conductive boards (Fig.3) and arranged in a cycle. The movement of the to-be-touch spheres creates a weightless and playful interaction (Fig.4) and two speakers at the centre of the installation begin to play sound phrases (Fig.6): “Am I weak or am I incapable? If...” followed by one of the 19 sensations, such as: “...my legs feel heavy” or “...my thoughts dissolve into fog.” In my design, I implemented the autosomatographical recordings and sensations that I felt during the performance. The interaction is intentionally subtle. The sound is quiet, and also sad, requiring listeners to step closer and engage with care. The hanging cords stretch (Fig.2) slightly on touch, creating gentle motion and soft resistance, adding to the sensory depth of the experience and the concept. The soundscape is composed of my intimate voice



Figure 3: conductive board and threads

recordings, captured in moments of bodily awareness and emotional recall. It is not meant to explain but to evoke. People can choose whether to stay outside, step into my cycle, or leave whenever they wish. Entering my hormonal cycle means stepping into the core of my inner world. It is a way to connect with me without my physical presence. Each small sensation sphere fits in one hand and reveals something deeply personal. Being surrounded by all the bouncing sensations, brings a lightness to the stories I share. The installation at the exhibition was designed as a space within a space, an opportunity environment that fosters closeness, exchange, and self-reflection. A small world of empathy, resonance, and reflection coming from the own sensations and proposing an alternative approach to menstruation literacy. The tactile interaction becomes a moment of embodied dialogue, making visible what is often concealed, and offering a space for both individual reflection and collective learning. This design language was chosen to mirror the often-unspoken nature of hormonal experience, felt but rarely shared. I create spaces for collective recognition, not just personal management,

4 Discussion

The installation *In My Cycle* offers a shift away from only data-driven or clinical representations of menstrual health toward more embodied, affective, and experiential forms. This relates to Campo Woytuk et. al. argument that only reading the body's data on mobile apps can risk perceiving our bodies as a stigmatist and disconnected separation from ourselves [5]. Wood suggests social change, better education and inclusion by breaking the concealment and gaining visibility of the suppressed gender [1]. Inspired by that I build my walkable hormonal cycle which creates space for self-reflection, visibility, and public engagement. The presented artefact addresses this concealment break by sharing my inner sensations, by offering a space for informal exchange and the opportunity of shared connection, what Mingus describes as *access intimacy* [2]. However, bodies are also different and it depends on the person how much

they want to share and know. According to Campo Woytuk et. al., technology should allow the user to define individually how much menstrual experience is preferred to be shared and how much to be kept in private [6]. The installation offers a space where people can also just consume, listen, and linger. However through sharing my own experiences, some of the deeply personal conversations sparked by the installation. This reaffirms Michie et al.'s suggestion that digital storytelling fosters empathy, amplifies diverse voices, and builds awareness through shared narratives[11]. I hope it invites a space of openness, understanding, and genuine connection with others.

Using my own body as both subject and artefact was an ambivalent experience, both empowering and deeply vulnerable. During the performance, I connected with my inner self in a way I had never experienced before. The sensations and the weight marked on my body revealed a new dimension of awareness. I wanted to translate these sensations into an experienced installation so others could relate to. As a designer, revealing personal work is familiar, but sharing my hormonal history and physical sensations in a public space felt at first uncomfortable. However, my feminist self understood that this response was shaped by cultural norms of concealment around menstruation, as described by Houppert [9]. Exhibiting the work created moments of both connection and distance. Some visitors engaged deeply, leading to powerful shared reflections; others remained silent, leaving me unsure of their responses. However, the externalized artefact gave me a sense of emotional distance and agency [7]. It acted as a protective boundary while enabling engagement, allowing me to share intimate knowledge without being fully exposed. In this way, the artefact became not just a communication tool, but a means of navigating vulnerability and connection. As Demir et. al. also describe having the possibility to distract oneself from the pain or of giving speech to a self-agentic artefact can lead to an empowering sensation of the owner [7].

5 Conclusion

This research has explored how soma design can serve as a methodology for the externalization of menstrual sensations, offering tools for self-reflection, body literacy, and informal public learning. Through the design and exhibition of *In My Cycle*, I have shown how embodied, autobiographical practice, rooted in memory, sensation, and performance, can inform the creation of interactive systems that speak to lived hormonal experiences. Furthermore, this project positions the body not as a subject of design but as an active agent in the design process [7]. Through performance, memory, and sensation, the installation becomes a space of shared reflection, where personal narratives can catalyze collective understanding. This highlights the potential of intimate, autobiographical design approaches to contribute meaningfully to feminist HCI, soma design, and public health discourse. By centering the menstruating self in design and making its sensations available to others through sound and touch, this project contributes to a more empathetic, embodied, and inclusive vision of interaction design.



Figure 4: interaction *In my Cycle*, engaging with the spheres



Figure 5: conductive elastic thread and wooden spheres

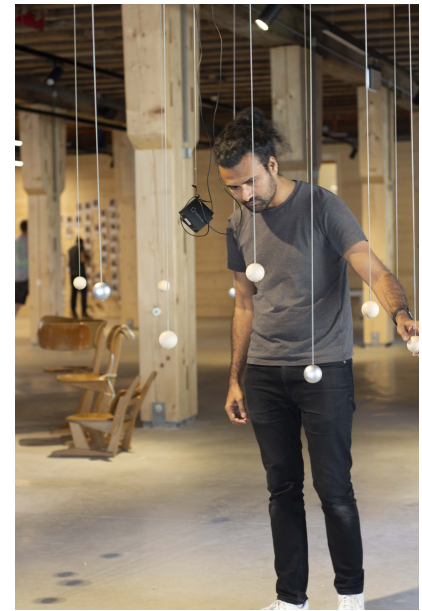


Figure 6: engaging and listening to the sound installation

Acknowledgments

I want to thank my supervision team Christopher Frauenberger, Verena Fuchsberger, Bernhard Maurer, for their guidance and Catharina Bauer and Christoph Döberl for the help I got by setting up the installation. This research project is part of the Paris-Lodron University of Salzburg and Salzburg University of Applied Sciences doctoral college doc.hci – Designing Meaningful Human-Technology Relations, supported by the Austrian Science Fund: DFG 12-N. This open access publication was supported by the Paris Lodron University of Salzburg.

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